

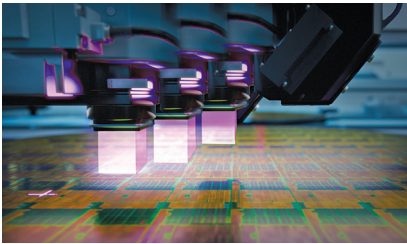
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Digital micromirror device

Engineers and manufacturers working on semiconductor packaging face the challenge of creating detailed circuit patterns without the time and cost of photomasks. Traditional mask-based lithography can be slow, expensive, and limiting when combining multiple



chips or technologies in a single package. Texas Instruments' DLP9910UV digital micromirror device addresses this by enabling maskless digital lithography. With 8.9 million pixels, sub-micron resolution, and a data rate of 110 gigapixels per second, it allows precise, scalable patterning without photomasks, helping manufacturers produce smaller, faster, and more energy-efficient systems at scale for applications like data centres and 5G, while streamlining production, lowering costs, and supporting multi-chip designs.

Texas Instruments (TI)
<https://www.ti.com>

AI-ready processor

Battery-powered edge devices struggle to run AI because traditional microcontrollers and NPUs are not optimised for neural networks, limiting tasks like voice recognition, keyword spotting, low-frequency computer vision, and sensing. Ambient Scientific's GPX10 Pro addresses this by using AI-native silicon that maps neural network operations directly to in-memory analogue compute blocks, cutting wasted cycles and overhead. It supports all common

neural network types and delivers up to 100x improvements in power, performance, and area over 32-bit MCUs,

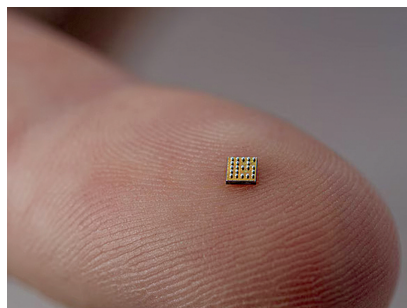


allowing developers to run AI on edge devices without draining batteries.

Ambient Scientific
<https://www.ambientscientific.ai>

Bluetooth chip

Many portable devices and IoT products face issues with size, power use, and responsiveness, forcing designers to compromise. Swatch Group's new Bluetooth chip addresses this by being the smallest on the market, allowing denser electronics and compact



designs, while using minimal energy to extend device autonomy. Its fast startup improves responsiveness and lifespan for devices like electronic beacons. Compatible with Bluetooth 5.0 and containing over 5 million transistors on 5mm², the chip can operate alone or with various sensors, giving system integrators the flexibility to add Bluetooth functionality to any portable

device with low power requirements.

CSEM
<https://www.csem.ch/en/contact>

Cellular-satellite IoT platform

Global IoT devices often lose connectivity when they move out of cellular coverage, disrupting applications like asset tracking, agricultural monitoring, and remote utilities. Traditional satellite solutions exist but are expensive, complex, and require multiple vendors, certifications, and platforms, increasing costs and deployment time.



Semtech and Skylo have solved this with the first complete device-to-cloud IoT solution combining terrestrial and satellite connectivity in one package. Using Skylo-certified HL7810 and HL7812 LPWA modules, the Smart Connectivity Premium SIM with integrated satellite access, and management via the AirVantage platform, devices automatically switch between networks to maintain continuous connectivity worldwide.

Semtech and Skylo
<https://www.semtech.com> and
<https://www.skylo.tech>

Supercapacitor-based DC UPS

Industrial systems, such as automation lines and telecommunications networks, often face power interruptions that cause downtime or data loss. Meanwhile, battery-based backup systems require maintenance and deteriorate in harsh environments. FEAS